



IDENTIDADES TRIGONOMÉTRICAS

Pitagóricas	Recíprocas	Suma a Producto
$\text{sen}^2(\theta) + \text{cos}^2(\theta) = 1$ $1 + \text{cot}^2(\theta) = \text{csc}^2(\theta)$ $\text{tan}^2(\theta) + 1 = \text{sec}^2(\theta)$	$\text{sec}(\theta) = \frac{1}{\text{cos}(\theta)}$ $\text{csc}(\theta) = \frac{1}{\text{sen}(\theta)}$ $\text{cot}(\theta) = \frac{1}{\text{tan}(\theta)}$	$\text{sen}(\alpha) + \text{sen}(\beta) = 2\text{sen}\left(\frac{\alpha + \beta}{2}\right)\text{cos}\left(\frac{\alpha - \beta}{2}\right)$ $\text{sen}(\alpha) - \text{sen}(\beta) = 2\text{cos}\left(\frac{\alpha + \beta}{2}\right)\text{sen}\left(\frac{\alpha - \beta}{2}\right)$ $\text{cos}(\alpha) + \text{cos}(\beta) = 2\text{cos}\left(\frac{\alpha + \beta}{2}\right)\text{cos}\left(\frac{\alpha - \beta}{2}\right)$ $\text{cos}(\alpha) - \text{cos}(\beta) = 2\text{sen}\left(\frac{\alpha + \beta}{2}\right)\text{sen}\left(\frac{\alpha - \beta}{2}\right)$
Cofusión	Reducción	Suma y Diferencia
$\text{sen}\left(\frac{\pi}{2} - \theta\right) = \text{cos}(\theta)$ $\text{sen}(\theta) = \text{cos}\left(\frac{\pi}{2} - \theta\right)$ $\text{sec}\left(\frac{\pi}{2} - \theta\right) = \text{csc}(\theta)$ $\text{sec}(\theta) = \text{csc}\left(\frac{\pi}{2} - \theta\right)$ $\text{tan}\left(\frac{\pi}{2} - \theta\right) = \text{cot}(\theta)$ $\text{tan}(\theta) = \text{cot}\left(\frac{\pi}{2} - \theta\right)$	$\text{sen}(-\theta) = -\text{sen}(\theta)$ $\text{cos}(-\theta) = \text{cos}(\theta)$ $\text{tan}(-\theta) = -\text{tan}(\theta)$ $\text{sec}(-\theta) = \text{sec}(\theta)$ $\text{cot}(-\theta) = -\text{cot}(\theta)$ $\text{csc}(-\theta) = -\text{csc}(\theta)$	$\text{sen}(\alpha + \beta) = \text{sen}(\alpha)\text{cos}(\beta) + \text{sen}(\beta)\text{cos}(\alpha)$ $\text{sen}(\alpha - \beta) = \text{sen}(\alpha)\text{cos}(\beta) - \text{sen}(\beta)\text{cos}(\alpha)$ $\text{cos}(\alpha + \beta) = \text{cos}(\alpha)\text{cos}(\beta) - \text{sen}(\alpha)\text{sen}(\beta)$ $\text{cos}(\alpha - \beta) = \text{cos}(\alpha)\text{cos}(\beta) + \text{sen}(\alpha)\text{sen}(\beta)$ $\text{tan}(\alpha + \beta) = \frac{\text{tan}(\alpha) + \text{tan}(\beta)}{1 - \text{tan}(\alpha)\text{tan}(\beta)}$ $\text{tan}(\alpha - \beta) = \frac{\text{tan}(\alpha) - \text{tan}(\beta)}{1 + \text{tan}(\alpha)\text{tan}(\beta)}$
Ángulo Doble	Reducción de Potencia	Producto a Suma
$\text{cos}(2\theta) = \text{cos}^2(\theta) - \text{sen}^2(\theta)$ $\text{sen}(2\theta) = 2\text{sen}(\theta)\text{cos}(\theta)$ $\text{tan}(2\theta) = \frac{2\text{tan}(\theta)}{1 - \text{tan}^2(\theta)}$	$\text{sen}^2(\theta) = \frac{1 - \text{cos}(2\theta)}{2}$ $\text{cos}^2(\theta) = \frac{1 + \text{cos}(2\theta)}{2}$ $\text{tan}^2(\theta) = \frac{1 - \text{cos}(2\theta)}{1 + \text{cos}(2\theta)}$	$\text{sen}(\alpha)\text{sen}(\beta) = \frac{1}{2}[\text{cos}(\alpha - \beta) - \text{cos}(\alpha + \beta)]$ $\text{cos}(\alpha)\text{cos}(\beta) = \frac{1}{2}[\text{cos}(\alpha - \beta) + \text{cos}(\alpha + \beta)]$ $\text{sen}(\alpha)\text{cos}(\beta) = \frac{1}{2}[\text{sen}(\alpha + \beta) + \text{sen}(\alpha - \beta)]$ $\text{cos}(\alpha)\text{sen}(\beta) = \frac{1}{2}[\text{sen}(\alpha + \beta) - \text{sen}(\alpha - \beta)]$